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DMFFT: improving the generation quality of diffusion models using fast Fourier transform

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In this study, we demonstrate the significant development potential of diffusion U-Net extraction features transferred to the frequency domain, opening up a new perspective for diffusion models and generating a new optimization idea for diffusion model-related research. The generating quality of Text-to-Image (T2I) or Text-to-Video (T2V) can be significantly enhanced by modifying the key indicators of U-Net frequency domain features. We first investigated the two types of modules in the sampling process, CrossAttnUpBlock and UpBlock, on U-Net, and then examined the effect of fine-tuning modules on U-Net feature extraction from backbone and lateral skip connections. Finally, it is decided to modify the feature extraction procedure of the CrossAttnUpBlock as the entrance to increase the overall diffusion generation quality. At the same time, employing the traditional Fourier transform method in image processing, the influence of frequency domain elements such as frequency, amplitude, and phase on image generation is investigated in the upsampling CrossAttnUpBlock of diffusion models, resulting in the method DMFFT, which can improve T2I or T2V quality without training or fine-tuning. During the experiments, the scaling factors for high frequency, low frequency, amplitude, and phase are adapted, generating a variety of characteristic performance results. A large number of experiments show that the proposed method, DMFFT, can significantly improve the semantic alignment, structural layout, color texture, and temporal consistency of images or videos generated by baseline models, as well as improving the artistry and diversity of the generation effect.

Diffusion Models[1–5], as the most advanced depth generation model family, are widely used in a variety of challenging image synthesis tasks and show application potential in a variety of fields. It mainly includes interdisciplinary applications in the fields of computer vision, natural language processing, temporal data modeling, multimodal modeling, medical image reconstruction, etc.[6]. The diffusion models belong to the probabilistic generation model, which gradually destroys data by adding noise to the original image, and then realizes sample generation by learning to reverse the process of adding noise. Traditional diffusion models usually operate directly in the pixel space, so powerful diffusion models often consume huge computational resources and lead to costly reasoning due to sequential evaluation. In order to achieve the generation quality and flexibility of Diffusion Models with limited computational resources, Latent Diffusion Model[5] apply the training and sampling process of diffusion model to the latent space of powerful pre-trained autoencoders. Thus, the superiority of diffusion models in image synthesis is maintained in limited computational resources. Stability AI transforms the theory of the latent diffusion models into an open source called Stable Diffusion, which can quickly generate high-quality images on consumer grade GPUs. Based on Stable Diffusion, researchers have carried out a series of rich and diverse researches and applications, which has led to the upsurge of Artificial Intelligence Generated Content (AIGC).

On the basis of the in-depth exploration and mining of the earlier Diffusion models theory[7–9], after the Stable Diffusion open source, researchers have focused on controlled image generation[10,11], example-based image editing[12], text-driven image editing[13], style transfer[14], image restoration[15], interactive image editing[16], and many other research areas to maximize the image generation potential of diffusion models. However, no matter what kind of field the diffusion models are applied to, the generated content that conforms to the text semantics, the exquisite visual effect of color texture, and the high-definition image resolution are the generation goals and improvement directions of all diffusion applications. Based on this common optimization goal, Susung Hong et al.[17] proposed a sampling quality optimization method based on self-attention mechanism in the latest study. ERNIE-ViLG 2.0[18] proposes to improve the Text-to-Image Diffusion Models with Knowledge-Enhanced Mixture-of-Denoising-Experts. SDXL[19] uses three times the size of the U-Net backbone, covers more

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attention modules and larger cross-attention contexts, and achieves an improvement in high-resolution image generation for Latent Diffusion Model. PriorGrad[20] uses data correlation adaptive prior conditions to improve the conditional denoising diffusion models. Most studies focus on improving attention mechanisms, conditional guidance, and network structure, all of which require specific computing resources and parameters. However, excessive increases in computing resources and parameters raise overall operating costs, and researchers will face a challenge in finding an effective balance between resource consumption and generation quality improvement. Inspired by the low resource occupation and high problem-solving efficiency of traditional mathematical methods, the researchers in this study intend to incorporate traditional mathematical methods into the current popular diffusion generation models in order to address the issue that improving generation quality requires a significant amount of computing resources.

Looking at the optimization research and application in the field of computer vision based on diffusion models, there are relatively few researches combining with traditional mathematical methods. In this study, Fourier transform method is introduced to improve the generation quality of diffusion models. Fourier Transforms, as one of the representatives of traditional mathematical methods, are applied to the field of computer vision to make use of their advantages in image processing. Fourier transform can be applied to many fields in computer vision, such as style transfer[21], image drawing[22], text detection of arbitrary shapes[23], network structure optimization[24], image coding[25], and image deblurring[26], etc. FreeU[27] investigates U-Net denoising applications in diffusion model, spectrum modulation is adopted in the Fourier domain to selectively reduce the low frequency components of lateral skip connections features, thereby alleviating the problem of excessive texture smoothing caused by enhanced denoising process. However, in FreeU, the backbone features are not transformed to Frequency domain, and the frequency domain's amplitude and phase are not applied further.

In this study, the backbone and the lateral skip connections of the diffusion model U-Net are comprehensively analyzed, and all of the key Fourier transform indicators such as high frequency, low frequency, amplitude and phase are applied to conduct in-depth research on the generation quality improvement of diffusion models. By the proposed method, we can selectively adjust some of the indicators to achieve the purpose of improving the generation quality. The proposed method provides a specific operational way for improving the generation quality. As shown in Fig. 1,the proposed method can find an effective balance between resource consumption and generation quality improvement. The main contributions are as follows:

- This work demonstrates the significant development potential of diffusion U-Net feature extraction results transformed into the frequency domain, which opens up new avenues of research for diffusion models.
- The impact of fine-tuning CrossAttnUpBlock and UpBlock in the diffusion U-Net sampling process on image feature extraction is thoroughly investigated, and the need to use CrossAttnUpBlock as the optimization operation object is clearly stated.
- After transforming the CrossAttnUpBlock feature extraction results in diffusion U-Net to the frequency domain, the effect of frequency, amplitude, and phase on the generation quality of T2I or T2V is investigated. Combined with the improvement of semantics, texture, color, clarity, etc., a diffusion model generation quality optimization method DMFFT based on Fourier transform is proposed. The DMFFT clarifies which elements of the generated results are improved by changing the scaling factors of frequency, amplitude, and phase. The proposed method also provides suggested value intervals for each scaling factor.
- By comparing the typical baselines, verify the experimental effect of DMFFT, and choose the fast Fourier transform (FFT) as the experimental method to reduce the time complexity and improve the computational efficiency.

Related work

Stable diffusion

Stable Diffusion is mainly based on the method of Latent Diffusion Models [5]. It mainly includes three core parts: perceptual image compression, latent diffusion and conditional mechanism. Perceptual image compression involves encoding an RGB image x into a latent representation $z = \epsilon(x)$, $z \in \mathbb{R}^{h\times w\times c}$ by an autoencoder ϵ, and

Fig. 1. Comparison of sample quality with and without DMFFT. The proposed DMFFT method combines Diffusion Models with Fourier transform. DMFFT costs: no training, no additional parameters, no increasing memory.

decoding the latent into an RGB image $\widetilde{x} = \mathcal{D}(z)$ by a decoder. The conditional mechanism mainly controls the generation process through text, semantic or other image-to-image conversion tasks, and introduces the control condition y into the U-Net cross-attention mechanism of the latent diffusion process to achieve conditional control. Latent diffusion, as the core part, mainly realizes the add noise and denoising process of the diffusion model, and the U-Net module of denoising process is the core of denoising. In the latent representation space, noise prediction network $\varepsilon_\theta(z_t, t)$ needs to be trained to realize the denoising process, and z_t can be obtained from ϵ during training. In the training of the latent diffusion model, z_t can be used to make the model learn in the latent representation space:

$$L_{LDM} = \mathbb{E}_{\epsilon(x),\varepsilon\sim\ N(0,1),t}\ [||\varepsilon - \varepsilon_\theta(z_t, t)||_2^2] \quad (1)$$

To implement the conditional guided image generation process, LDM introduces a domain-specific encoder τ_θ for preprocessing condition y :

$$L_{LDM} = \mathbb{E}_{\epsilon(x),y,\varepsilon\sim\ N(0,1),t}\ [||\varepsilon - \varepsilon_\theta(z_t, t, \tau_\theta(y))||_2^2] \quad (2)$$

U-Net structure in diffusion model

The U-Net neural network structure was first proposed by Olaf et al.[28] and is mainly used for medical image segmentation. The simple, efficient and stable network structure is not only widely used in the field of image segmentation, but also applicable to image denoising. In the structure of the diffusion model[5], U-Net is the core module, it mainly performs iterative denoising of Gaussian noise matrix in diffusion cycle. In diffusion denoising process, text and timesteps guide each noise prediction, after iterative denoising, finally, the random Gaussian noise matrix is transformed into the latent feature of the image. The U-Net module in the diffusion model is a key factor in determining the quality of the generated results such as T2I and T2V.

Matthew et al., in their study on visual understanding convolutional networks[29], pointed out that the semantic information and spatial information contained in the features extracted by each layer of convolutional neural networks are different: the features learned by the lower layer are mainly spatial information such as color and edge; The features learned by the higher convolutional layer are more distinctive and discriminative semantic features. Therefore, in the U-Net module of Stable Diffusion structure, the upsampling stage of U-Net fuses the low-level feature map of the corresponding position by using lateral skip connection, so as to realize the fusion of low-level feature and high-level feature. The main structure of the core module U-Net in the diffusion model is shown in Fig. 2.

Fourier transform

The Fourier transform, a mathematical operation that decomposes a signal into its component frequency components, is widely used in signal processing, as well as for tasks such as image recognition, object detection, and image compression. Fourier transform transforms images into frequency domain to detect visual features such as edges and shapes. In machine learning, Fourier transform can be applied to convolutional neural networks to improve the performance of image classification and recognition by detecting certain frequency components in images. Similarly, the data can be enhanced by changing the frequency component of the image.[30,31]et al. show that in convolutional neural networks, filtering high and low frequency noise by Fourier transform will have an impact on image contour, shape and semantic features.

For time domain signal $f(t)$, transform it into the frequency domain by Fourier transform, which can be expressed as:

$$F(\omega) = \int_{-\infty}^{\infty} f(t)\, e^{-i\omega t} dt \quad (3)$$

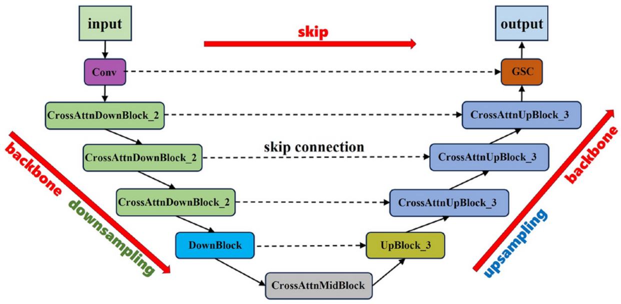


Fig. 2. U-Net Structure in Diffusion Model. In U-Net structure, CrossAttnDownBlock_2 means downsampling by two layers of ResNetBlock and Spatial Transformer. CrossAttnUpBlock_3 means sampling by three layers of ResNetBlock and Spatial Transformer. DownBlock consists of two layers of ResNetBlock. UpBlock 3 consists of three layers of ResNetBlock. CrossAttnMidBlock is composed of ResNetBlock + Spatial Transformer + ResNetBlock.

For signal a in the frequency domain, its corresponding inverse Fourier transformation into the time domain can be expressed as:

$$f\left(t\right)=\frac{1}{2\pi}\int_{-\infty}^{\infty}F\left(\omega\right)e^{i\omega t}d\omega \tag{4}$$

By converting the Fourier series in exponential form to the Fourier series in trigonometric function form, the periodic signal $f\left(t\right)$ is composed of a series of sine and cosine signals, which can be expressed as:

$$f\left(t\right)=a_0+\sum_{n=1}^{\infty}\left[a_n cos\left(n\omega_1 t\right)+b_n sin\left(n\omega_1 t\right)\right] \tag{5}$$

where the first harmonic frequency $\omega_1=2\pi/T_1$, DC component $a_0=\frac{1}{T_1}\int_{t_0}^{t_0+T_1}f\left(t\right)dt$, a_n is the amplitude of the cosine component, $cos\left(n\omega_1 t\right)$ represents the waveform of the cosine component, b_n represents the amplitude of the sine component, $sin\left(n\omega_1 t\right)$ represents the waveform of the sine component,$n\omega_1$ is called the NTH harmonic, according to the sine and cosine auxiliary Angle formula, $f\left(t\right)$ can be transformed into the following two forms:

A periodic signal consisting of a constant and sinusoidal signal:

$$f\left(t\right)=d_0+\sum_{n=1}^{\infty}d_n sin\left(n\omega_1 t+\theta_n\right) \tag{6}$$

A periodic signal consisting of a constant and cosine signal:

$$f\left(t\right)=c_0+\sum_{n=1}^{\infty}c_n cos\left(n\omega_1 t+\varphi_n\right) \tag{7}$$

where $a_0=d_0=c_0$, amplitude $d_n=c_n=\sqrt{a_n^2+b_n^2}$, phase $\theta_n=arctan\left(\frac{a_n}{b_n}\right)$, $\varphi_n=arctan\left(\frac{b_n}{a_n}\right)$.

The calculated signal $f\left(t\right)$ in Eqs. 3 is a continuous spectrum, but in the process of computer acquisitionïijŇthe value can be collected is the discrete sampling value of the continuous signal. Therefore, the discrete Fourier transform (DFT) method should be used to discretize the continuous time domain signal:

$$X\left(k\right)=\sum\nolimits_{n=0}^{N-1}x\left(n\right)e^{-i\frac{2\pi}{N}kn},\ 0\le k\le N-1 \tag{8}$$

where, the result in the frequency domain is a continuous periodic function with period 2π, then the time limit signal is sampled in the frequency domain at intervals of $\frac{2\pi}{N}$, $0\le k\le N-1$ is a frequency domain period, and $\frac{2\pi}{N}k$ corresponds to ω in the Fourier transform. Obviously, the complexity of computing one term $X\left(k\right)$ is $O\left(N\right)$, and the complexity of computing all the terms will reach $O\left(N^2\right)$.

In the calculation process, due to the high complexity of discrete Fourier transform, Fast Fourier transform (FFT) is introduced to further simplify the computational complexity and improve the efficiency. FFT is an algorithm that simplifies DFT calculation using divide-and-conquer idea.

For polynomials $f\left(x\right)=\sum\nolimits_{i=0}^{n}a_i x^i$, $g\left(x\right)=\sum\nolimits_{i=0}^{n}b_i x^i$ define their product *fg* as:

$$fg\left(x\right)=\left(\sum\nolimits_{i=0}^{n}a_i x^i\right)\left(\sum\nolimits_{i=0}^{n}b_i x^i\right) \tag{9}$$

We can calculate the coefficients of each term of this product by the complexity of $O\left(N^2\right)$, but FFT can calculate the product by the time complexity of $O\left(N\log N\right)$. Reduce time complexity with FFT, the discrete Fourier transform executed on the computer can be realized quickly.

Methods

U-Net, the diffusion model's core module, has a significant impact on the quality of the generated results, particularly the proportion of feature extraction results from backbone and lateral skip connections in the upsampling process, which directly affects the semantic expression of the generated results. In this section, the influence of the main blocks of the two branches on feature extraction is thoroughly investigated and the feature extraction results are transformed to the frequency domain for feature adjustment, realizing the feature extraction optimization process of resampling. Fig. 3 depicts the frequency, amplitude, and phase results obtained by transforming the image into the frequency domain.

The method proposed in this study primarily depends on an investigation of the T2I generation process of the text 'A cat riding a motorcycle.', and is verified by other T2I tests. In the method analysis statement, b represents the backbone feature scaling factor, and s represents the lateral skip connections feature scaling factor. 'Lateral skip connections' is abbreviated as 'skip'. 'backbone branch' is abbreviated as 'backbone'. The values in the figure represent the scaling factor, which is set in the range of [0.0, 1.9], and the experimental results in this range are relatively effective. The range shown in each figure covers the quality change process of the generated

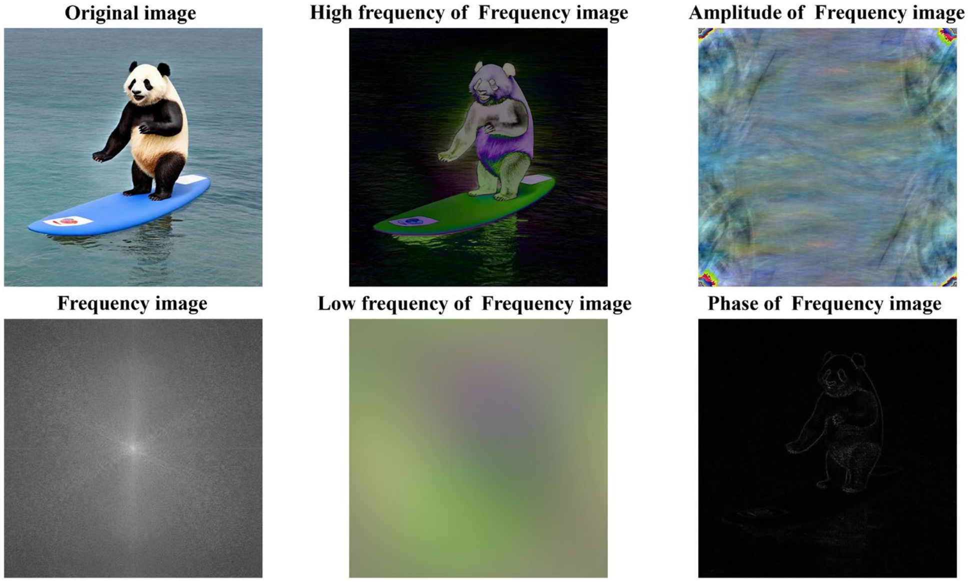


Fig. 3. Image transforming to frequency domain. High frequency, low frequency, amplitude, and phase can all be obtained through the Fourier transform. The high frequency corresponds to the gray scale component that changes quickly in the image, the low frequency corresponds to the gray scale component that changes slowly in the image, the amplitude represents the total space occupied by the frequency, and the phase represents the average position of the space corresponding to the frequency. After adjusting the frequency, amplitude, and phase of the image frequency domain, the adjusted result can be obtained through the inverse Fourier transform.

results. Because the change process of each frequency domain component is different, the displayed change interval is different.

Block fine-tuning

U-Net's upsampling process in the diffusion model is composed of stacked CrossAttnUpBlocks and UpBlocks. The two types of blocks combine the lateral skip connection of the corresponding position during feature extraction to achieve feature fusion at both high and low levels. Now, discuss the influence of the proportion of extracted features of the two types of blocks on the final generated results.

UpBlock fine-tuning

Figure 4 shows that while the UpBlock backbone feature scaling factor b is gradually increased, the generated images change slowly, and the majority of the images are consistent with the text, so semantic integrity can be maintained when $b = 1$. When the UpBlock skip feature scaling factor grows, the resulting image's background and text consistency weakens. When s is gradually reduced, text consistency improves, but background optimization is not as obvious. Through comprehensive analysis, UpBlock fine-tuning has no obvious effect on the improvement of the generated results. Furthermore, the UpBlock feature extraction process serves as the foundation for CrossAttnUpBlock to get feature images, therefore the UpBlock feature extraction results can be left unchanged.

CrossAttnUpBlock fine-tuning

As illustrated in Fig. 5, as the CrossAttnUpBlock backbone feature scaling factor b increases, the semantics of the generated image align with the text, and the generated image is more clear, but the background texture gradually blurs from $b = 1.3$. When the skip feature scaling factor s gradually increases, the semantic of the generated image weakens and the background texture becomes more visible. When s is small, the semantic expression is relatively better. To achieve high-quality generated images throughout the experiment process, the scaling factors of CrossAttnUpBlock backbone and skip can be valued in the interval [1.1, 1.3] and [0.5, 0.6] respectively.

The above fine-tuning blocks reveal that CrossAttnUpBlock has a more obvious effect on image generation, and the backbone is more focused on semantic expression, while the skip is more focused on color and texture expression. Therefore, the backbone and skip of CrossAttnUpBlock could be thoroughly analyzed. Create scale-adaptive block fine-tuning formulas to improve the quality of the generated results:

$$F_m^{'} = bF_m \quad F_n^{'} = sF_n \tag{10}$$

where F_m represents the CrossAttnUpBlock backbone feature map, F_n represents the CrossAttnUpBlock skip feature map, b is the CrossAttnUpBlock backbone feature scaling factor, s is the CrossAttnUpBlock skip feature

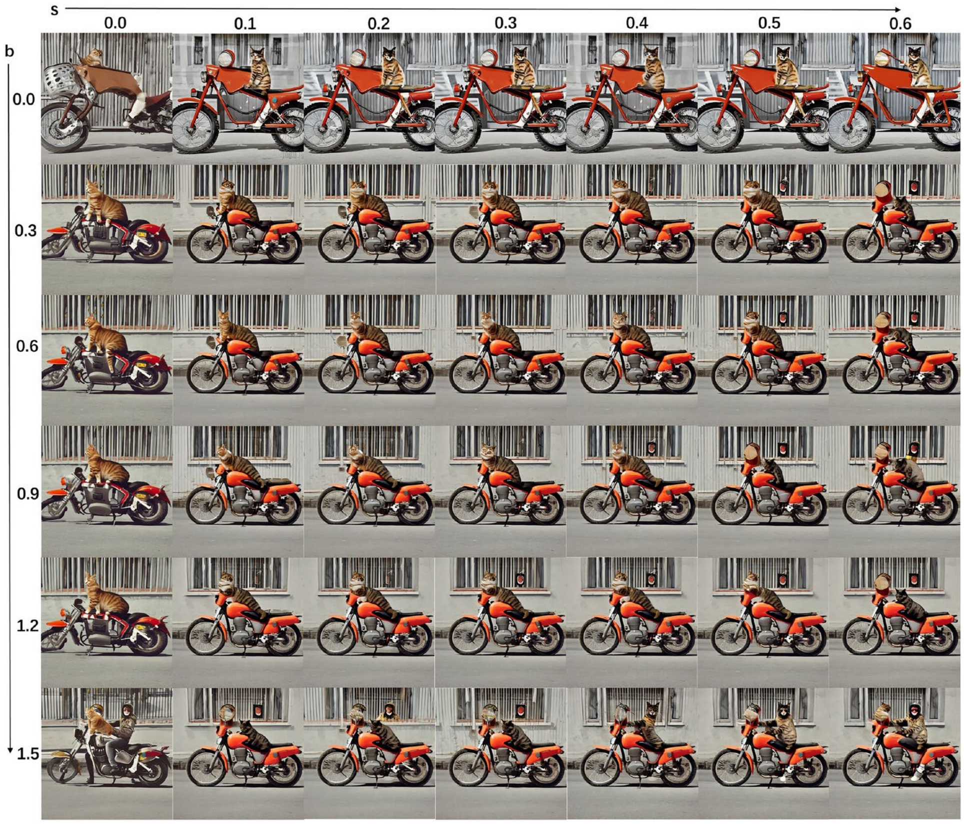


Fig. 4. Scaling factors fine-tuning of backbone and skip features for UpBlock.

Fig. 5. Scaling factors fine-tuning of backbone and skip features for CrossAttnUpBlock.

scaling factor, and $F_m^{'}$ represents the feature map after scaling the backbone factor. $F_n^{'}$ represents the feature map after scaling the skip factor.

Based on the core role of the diffusion model U-Net, as well as the impact of the CrossAttnUpBlock backbone and skip in the upsampling process on image quality, an optimal network for the sampling process on the diffusion model U-Net is built. This network transforms the feature extraction results of CrossAttnUpBlock during the upsampling process to the frequency domain. That is, the backbone branch and skip branch of the CrossAttnUpBlock upsampling process are respectively transformed to the frequency domain. The image features' all frequency domain components are scaled in the frequency domain to achieve optimal sampling. The scaled frequency domain results are transformed into the time domain, obtaining the optimal generated results. The structure diagram of the upsampling optimization network on the diffusion model U-Net is shown in Fig. 6.

High and low frequency scaling in the Fourier transform

This section analyzes the impact of the high and low frequencies of each branch of CrossAttnUpBlock on the upsampling process. Fig. 7 illustrates that increasing the proportion of high frequency features improves image semantics and optimizes color and brightness. When the proportion of high-frequency features decreases, the image's contour softens and the boundary blurs. When the proportion of low frequency features decreases, image semantics improve, clarity improves, and the image's color and texture are adjusted. As the proportion of low frequency features increases, the contour of the image softens and the boundary blurs. It is worth noting that when the proportion of high frequency components in the skip increases, the background texture appears uneven, but color brightness and semantics improve. The generation impact is relatively good when the scaling factors for high and low frequency features are in the intervals [1.2, 1.4] and [0.0, 0.8], respectively. Fig. 8 shows how the Fourier high frequency and low frequency feature scaling method affects the resulting image.

The preceding research shows that applying the Fourier transform to the upsampling process of the U-Net can improve the quality of the generated image. Therefore, the effects of high frequency and low frequency feature scaling should be incorporated into eqs. 10:

$$F_m^{'} = b \times IFFT(FFT\,(F_m) \odot (M_h \times \theta_{mh}) + FFT\,(F_m) \odot (M_l \times \theta_{ml})) \tag{11}$$

$$F_n^{'} = s \times IFFT(FFT\,(F_n) \odot (N_h \times \theta_{nh}) + FFT\,(F_n) \odot (N_l \times \theta_{nl})) \tag{12}$$

where M_h and M_l represent the backbone's high and low frequency masks, while N_h and N_l represent the skip's high and low frequency masks, θ_{mh} and θ_{ml} represent the scaling factors of the backbone's high and low frequency features, θ_{nh} and θ_{nl} represent the scaling factors of the skip's high and low frequency features.

Amplitude and phase scaling in Fourier transform

In signal processing, by controlling the amplitude and phase of different frequency components of Fourier transform, a variety of signals with different waveforms can be synthesized. Similarly, in image processing, phase and amplitude can be utilized to describe image features. Phase can reflect an image's texture and structure,

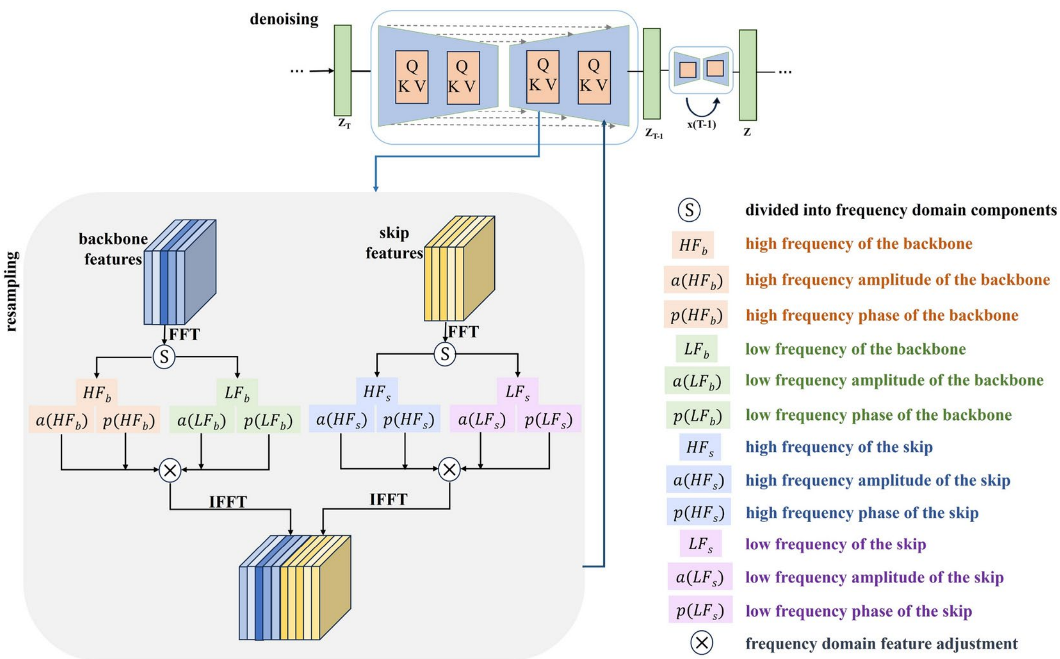


Fig. 6. Structure of upsampling optimization network on diffusion model U-Net. FFT represents the fast Fourier transform, IFFT represents the inverse fast Fourier transform.

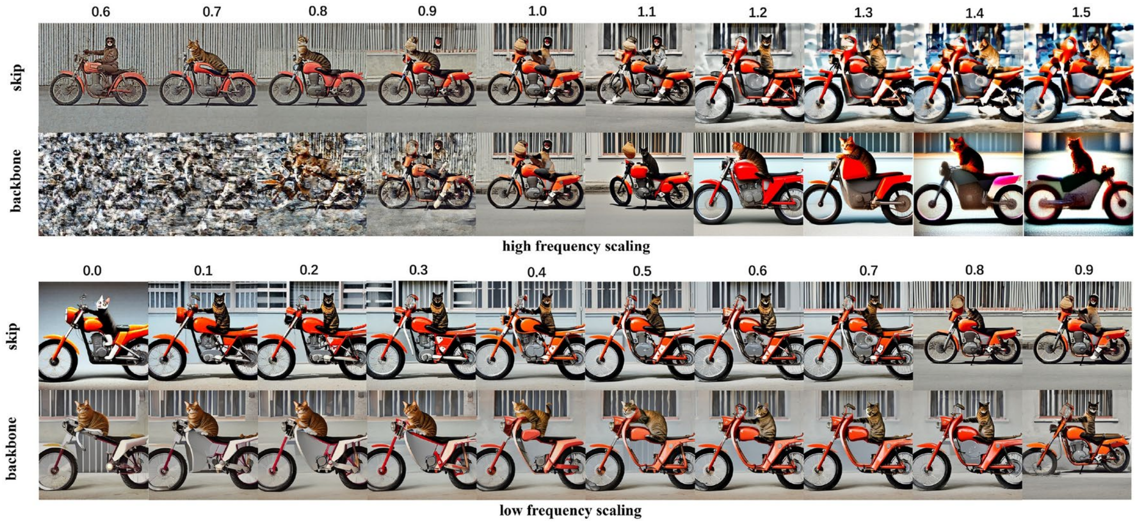


Fig. 7. Scale the proportion of high frequency and low frequency features.

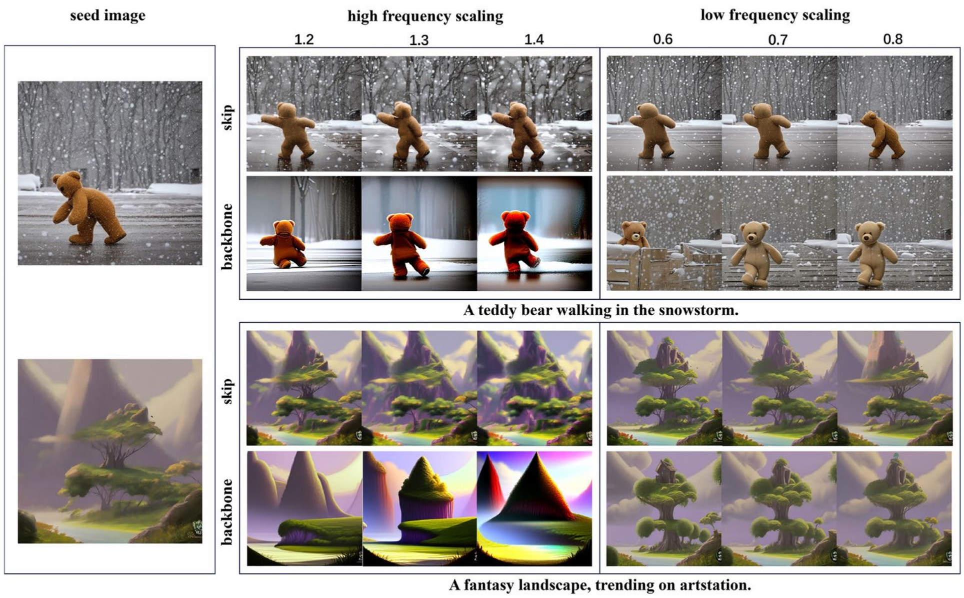


Fig. 8. The impact of Fourier transform high frequency and low frequency feature scaling on T2I generating results. Based on the seed image, the high frequency and low frequency features are scaled to the suggested interval by the Fourier transform, and the T2I results are shown in the figure.

whereas amplitude can reflect its brightness and contrast. The impact of scaling the amplitude and phase of different frequency components on T2I upsampling process is discussed below.

High frequency unchanged, low frequency amplitude and phase analysis

The high frequency features of the backbone and skip of CrossAttnUpBlock remain unchanged, investigate the impact of low frequency amplitude and phase scaling. Fig. 9 shows that as the low frequency amplitude scaling factor is reduced, the generated image quality improves and when the scaling factor is smaller than 0.7, the generated results are relatively good.

As illustrated in Fig. 10, increasing and decreasing the low frequency phase proportion in both the backbone and skip features can lead to quality improvements. Higher semantic, color, texture, and clarity generation effects can be obtained when the low frequency phase scaling factors of the backbone and skip features are not in the interval [0.8, 1.2]. The experimental verification results are shown in Fig. 11.

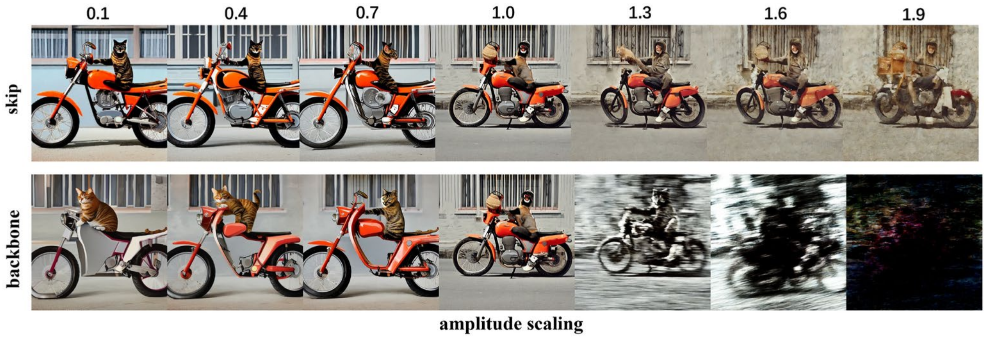


Fig. 9. The high frequency features remain unchanged, the impact of low frequency amplitude scaling.

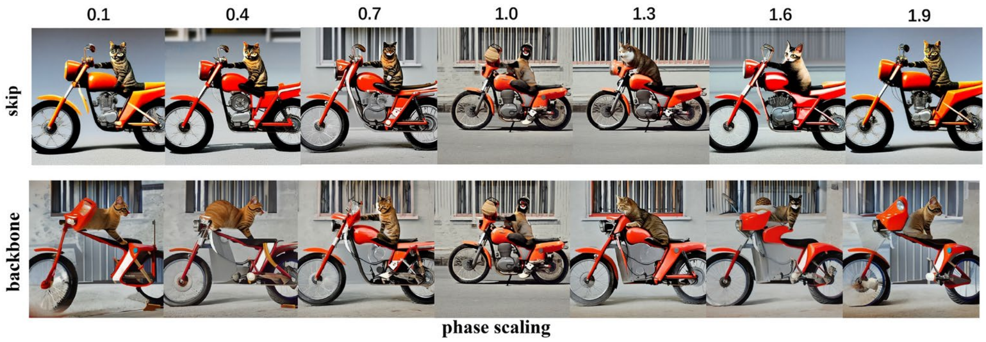


Fig. 10. The high frequency features remain unchanged, the impact of low frequency phase scaling.

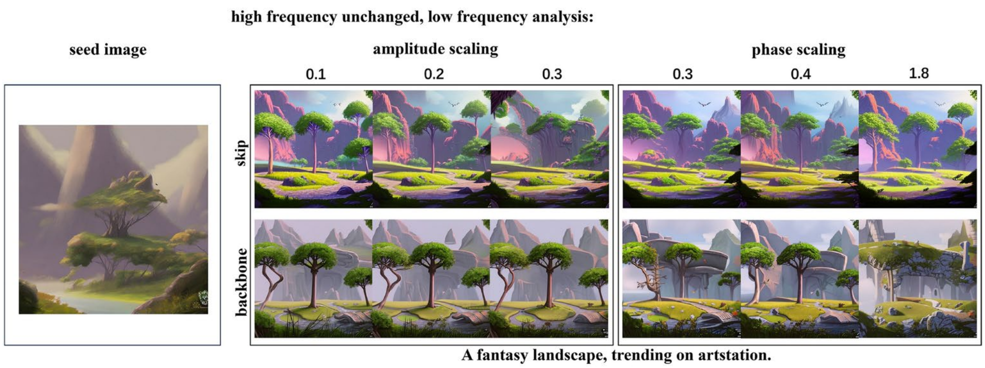


Fig. 11. The effect verification of low frequency amplitude and phase scaling. With low frequency amplitude and phase scaling, the T2I results achieve changes in position or layout of the semantic content while aligned with the text.

Low frequency unchanged, high frequency amplitude and phase analysis

As illustrated in Fig. 12, the low frequency features of CrossAttnUpBlock's backbone and skip remain unchanged. When the high frequency amplitude scaling factors of the backbone and skip are increased, the generated images have better color and semantic expression, but the skip results show patches in the background, whereas the backbone results show more prominent performance. When the high frequency amplitude scaling factors of the backbone and skip features are between [1.2, 1.4], all of the results are relatively satisfactory.

As illustrated in Fig. 13, while the low frequency features of the backbone and skip remain unchanged, the generated results can be aligned with the text by increasing or decreasing the high frequency phase scaling factor of the skip. When the scaling factor is less than 0.9 or larger than 1.0, the generation results are acceptable but the overall quality is poor. When the high frequency phase scaling factor of the backbone is increased or decreased, none of the obtained results improve, so the experimental process can be unchanged. As illustrated in Fig. 14, when the high frequency amplitude or phase scaling factor is scaled while the low frequency features remain unharmed, the generation quality is further verified.

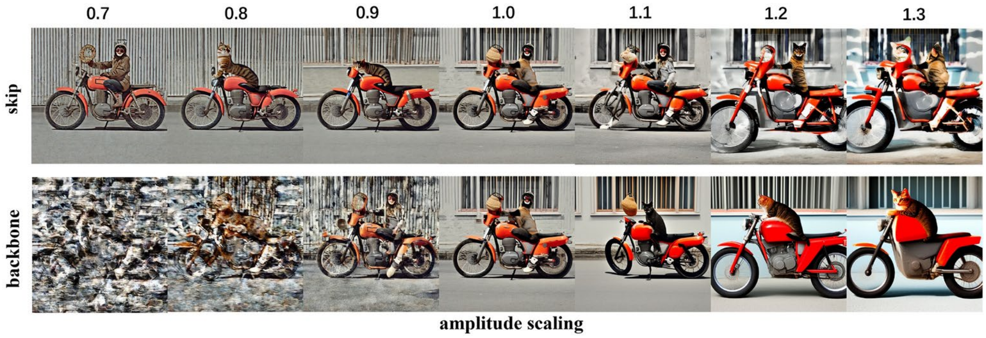


Fig. 12. The low frequency features remain unchanged, the impact of high frequency amplitude scaling.

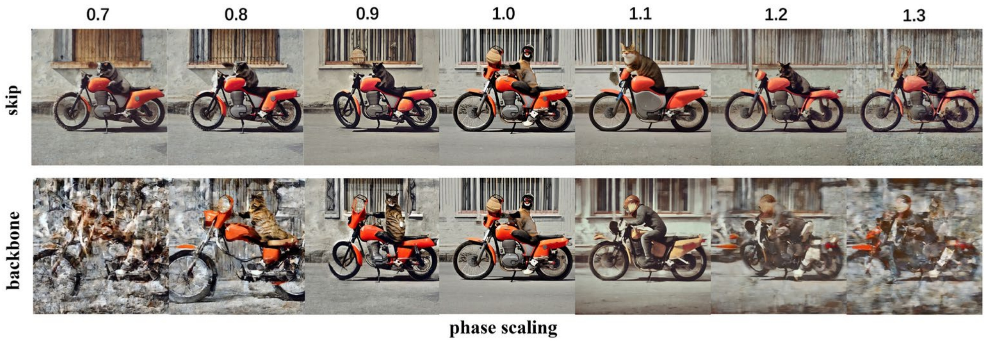


Fig. 13. The low frequency features remain unchanged, the impact of high frequency phase scaling.

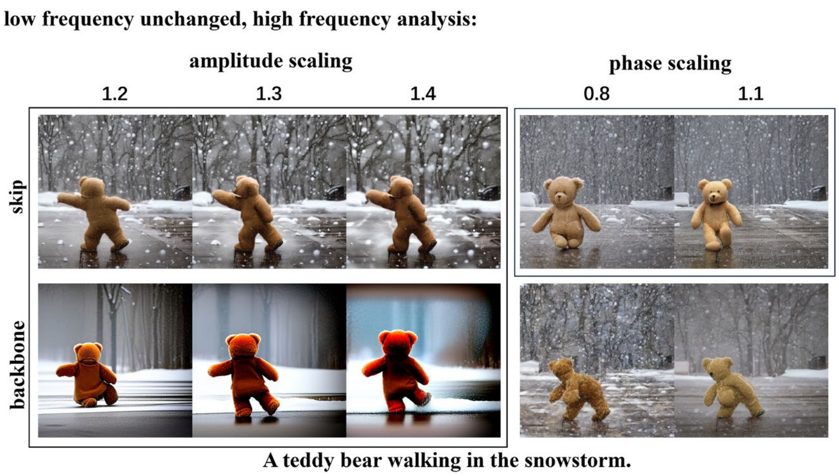


Fig. 14. The effect verification of high frequency amplitude and phase scaling. Backbone phase scaling has a small optimization effect on the generated results, so it is not recommended.

In conclusion, the impact of Fourier transform amplitude and phase on image generation can be incorporated into the Fourier transform high frequency and low frequency proportional scaling methods to construct a comprehensive optimization strategy DMFFT:

$$\begin{aligned} F_m^{ha} &= abs\left(FFT\left(F_m\right) \odot \left(M_h \times \theta_{mh}\right)\right) \\ F_m^{la} &= abs\left(FFT\left(F_m\right) \odot \left(M_l \times \theta_{ml}\right)\right) \\ F_m^{hp} &= angle\left(FFT\left(F_m\right) \odot \left(M_h \times \theta_{mh}\right)\right) \\ F_m^{lp} &= angle\left(FFT\left(F_m\right) \odot \left(M_l \times \theta_{ml}\right)\right) \\ F_m^{'} &= b \times IFFT(F_m^{ha} \times a_{mh} + F_m^{la} \times a_{ml} + F_m^{hp} \times p_{mh} + F_m^{lp} \times p_{ml}) \end{aligned} \tag{13}$$

$$\begin{aligned} F_n^{ha} &= abs\left(FFT\left(F_n\right) \odot \left(N_h \times \theta_{nh}\right)\right) \\ F_n^{la} &= abs\left(FFT\left(F_n\right) \odot \left(N_l \times \theta_{nl}\right)\right) \\ F_n^{hp} &= angle\left(FFT\left(F_n\right) \odot \left(N_h \times \theta_{nh}\right)\right) \\ F_n^{lp} &= angle\left(FFT\left(F_n\right) \odot \left(N_l \times \theta_{nl}\right)\right) \\ F_n^{'} &= s \times IFFT(F_n^{ha} \times a_{nh} + F_n^{la} \times a_{nl} + F_n^{hp} \times p_{nh} + F_n^{lp} \times p_{nl}) \end{aligned} \tag{14}$$

The result of the fast Fourier transform is in complex number form, so 'abs' is used to describe the process of taking the amplitude (the square root of the real and imaginary parts of the complex number). The phase is also converted from the complex number form to the corresponding angle, so in the formula phase is represented by 'angle'. In the formula, F_m^{ha} and F_m^{la} represent the amplitudes of the high and low frequencies of backbone respectively, while a_{mh} and a_{ml} represent the corresponding scaling factor. F_m^{hp} and F_m^{lp} are the phases of the backbone's high and low frequencies, respectively p_{mh} and p_{ml} are the corresponding scaling factors. In the formula, F_n^{ha} and F_n^{la} represent the amplitudes of the high and low frequencies of skip, while a_{nh} and a_{nl} represent the corresponding scaling factor, respectively. F_n^{hp} and F_n^{lp} are the phases of the high and low frequencies of skip, respectively, and p_{nh} and p_{nl} are the corresponding scaling factors. When only phase and amplitude scaling is analyzed, the values of θ_{mh}, θ_{ml}, θ_{nh}, θ_{nl}, b and s can be set to 1 accordingly.

The analysis process of the DMFFT method above makes it clear that Fourier transform can improve the quality of the sampling results of the diffusion model, but in the specific application process, the scaled objects can be selected purposefully according to different experimental objectives. Scaled objects can be individually selected for frequency, phase or amplitude, it is not recommended to choose too much, otherwise it may cause excessive smoothness. The proportional scaling of the high frequency (HF), low frequency (LF), amplitude and phase of the backbone and the skip has different degrees of influence on each part of the generated image, as illustrated in Table 1.

As shown in Table 1, the same frequency domain component has the same value interval on the backbone and skip branches. However, based on the comparison effect of Figs. 8, 11 and 14 in the method section, it is discovered that the backbone and skip branches have distinct effects on the realism, clarity, special-effect, and modelization of the obtained results for the same frequency domain component. As illustrated in Fig. 8, when the scaling factor of the backbone branch's high-frequency components falls within the suggested value interval, the generated results resemble movie and television special effects or modeling effects. In the same value range, the skip branch generates a more realistic effect. Similarly, the high frequency amplitude of the backbone in Fig. 14 also shows the effect of the movie and television special effects. Furthermore, as illustrated in Fig. 11, whether the amplitude or phase of low frequency are in the same value interval, the backbone and skip branches generate significantly different effects. The diversity of the above generation results gives researchers numerous

Branch	Scaling factor	Value interval	Text alignment	Structural layout	Color Texture	Temporal Consistency
skip	high frequency	[1.2, 1.4]	✓	✓		
	low frequency	[0.0, 0.8]	✓	✓	✓	
	LF amplitude	[0.0, 0.7]	✓	✓	✓	✓
	LF phase	[0.0, 0.7] or [1.3, 1.9]	✓	✓	✓	✓
	HF amplitude	[1.2, 1.4]	✓	✓		
	HF phase	[0.0, 0.8] or [1.1, 1.2]	✓	✓		
backbone	high frequency	[1.2, 1.4]	✓	✓	✓	✓
	low frequency	[0.0, 0.8]	✓	✓		
	LF amplitude	[0.0, 0.7]	✓	✓	✓	
	LF phase	[0.0, 0.7] or [1.3, 1.9]	✓	✓	✓	
	HF amplitude	[1.2, 1.4]	✓	✓	✓	
	HF phase					

Table 1. The effect of high frequency, low frequency, amplitude, and phase scaling on generation results. The value interval is the suggested interval obtained by the proposed method.

generation effect options. In actual applications, the frequency domain components that fulfill the requirements of the generation target can be chosen for adjustment. The proposed method can be customized to some extent.

Experiments

To validate the generation effect of the Fourier transform-based diffusion model generation quality improvement method (DMFFT) proposed in this study, this section focuses on the two primary application fields of diffusion models, T2I and T2V, and choose representative baselines to conduct qualitative and quantitative studies. Further investigate the improvement of key elements which affect the quality of image and video. It is worth noting that the method proposed in this study will make the generated image or video focus on different improvements by adjusting scaling factors for high frequency, low frequency, amplitude and phase in the Fourier frequency domain.

The implementation of this study is based on the official published code for each comparison baseline and follows the optimal hyperparameter settings for each baseline (e.g., timesteps, guidance scale, and resolution). The method proposed in this study does not add additional model parameters, and requires no training during the experiments. Based on the idea of Fourier transform, this study chooses the fast Fourier transform (FFT) as the experimental method to reduce the time complexity and improve the computational efficiency. For the setting of scaling factors, we describe them in the specific comparison experiments. All experiments are conducted on a NVIDIA RTX 4090 GPU.

Text-to-Image (T2I)

T2I is a primary application field of Diffusion models. In this study, Stable Diffusion[5], FreeU[27] and SCALECRAFTER[32] are selected as the baselines for experimental comparison of T2I. FreeU introduced spectrum modulation into the process of diffusion model denoising, which alleviated the problem of excessive texture smoothing caused by enhanced denoising. SCALECRAFTER achieves high resolution T2I generation of diffusion models.

Improve text alignment to avoid structural semantic distortion

The proposed method (DMFFT) has excellent performance in improving the semantics of T2I. The first line of Fig. 15 compares the performance of SD, FreeU, and DMFFT in identifying face details. The results provided by SD and FreeU, as shown in the yellow box, show anomalous morphological features in the nose and facial transition area, different eye sizes, and hand textures on clothing. All the above anomalies have been improved in the DMFFT experiment results, and the results generated by DMFFT are more clearly. The results of the three methods on human posture (second line) are compared. As shown in the white box, the head and hands in the SD generated results are abnormally blurred, and the arms and legs in the FreeU generated results do not conform to the normal structure and posture of the human body. DMFFT has better performance in head morphology and body posture expression. In comparison to the results of the third experiment, the 3D effect

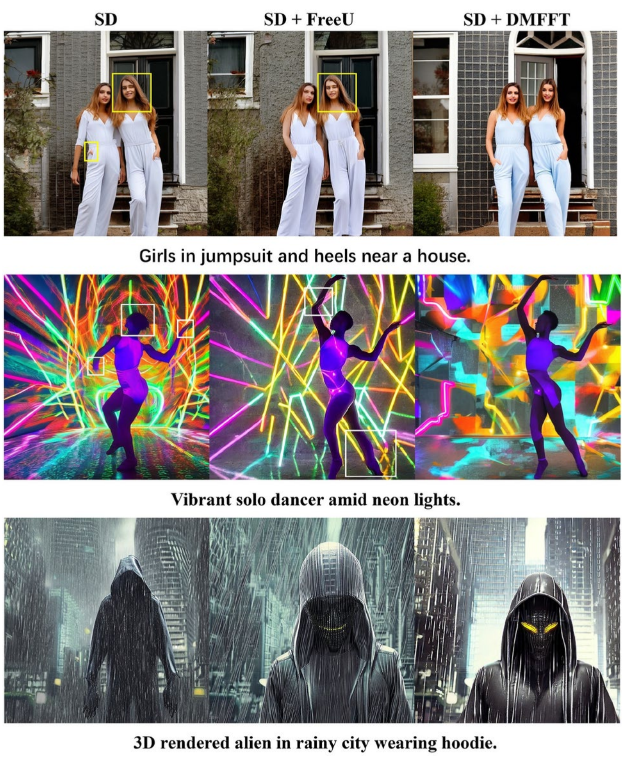


Fig. 15. Comparison of text alignment of SD, SD + FreeU, SD + DMFFT.

of the SD experiment is not obvious, and the semantic expression of foreground and background is extremely fuzzy. In the FreeU experiment results, the foreground information expresses the 3D effect, but the background information is still fuzzy. It is found that DMFFT can improve and enhance the 3D character effect as well as the semantic representation of the foreground background, which is superior than other comparable baselines.

The images generated by the DMFFT in Fig. 15 correspond to the following scaling factors: 1. The low frequency scaling factor of skip is set to 0.4. 2. The high frequency is unchanged, the backbone's low frequency phase scaling factor is set to 0.3. 3. The high frequency is unchanged, the backbone's low frequency amplitude scaling factor is set to 1.1, and the skip's low frequency amplitude scaling factor is set to 0.2. Here, the scaling factors of the two branches are mixed and adjusted, and the comprehensive value of the two factors conforms to the recommended interval. The above analysis shows that adjusting the low frequency scaling factor is critical to maintaining the semantics of the generated image.

Fig. 16 compares experimental results to SCALECRAFTER, which focuses on high-resolution T2I generation. As the red text description shows, in the case of text alignment with "arctic tundra", The results generated by SCALECRAFTER do not reflect the geological features of arctic tundra, whereas DMFFT achieves text alignment and improves image color and texture detail. For the text alignment of "contains sailing boat", the image generated by SCALECRAFTER does not conform to the normal visual Angle relationship between sailing boat and sea water. Instead, DMFFT maintains alignment with the text and conforms to the visual perspective. The images generated by the DMFFT in Fig. 16 correspond to the following scaling factors: 1. The skip's high frequency scaling factor is set to 1.3. 2. The high frequency is unchanged, the skip's low frequency phase scaling factor is set to 1.3.

Enhance visual effects and improve image quality

On the basis of text alignment, the DMFFT method proposed in this study also shows exceptional performance in visual effect enhancement. As shown in Fig. 17, the resolution of a single image is 768*768. At the same resolution, DMFFT's generation effect has a greater visual impact than SD and FreeU, and DMFFT can show character details more vividly and elegantly, character images are more three-dimensional, and text alignment is more visible. In Fig. 17, the following scaling factors are set respectively for DMFFT from top to bottom: 1. The high frequency remains unchanged, while the skip's low frequency phase scaling factor is set to 0.2. 2. The backbone's high frequency scaling factor is set to 1.2. The value of the first scaling factor conforms to the suggested value interval of Fig. 10, and the second conforms to the high-frequency scaling variation in the second row of Fig. 7, which further verifies the influence of scaling factor variation on the generated results.

For SCALECRAFTER's high-resolution (2048*2048) results, the DMFFT method can also improve the color textures, character details, and 3D effects. In the DMFFT results, as shown in the first row of Fig. 18, the layout in the traditional building is more consistent with the decoration of the teahouse, and the sky and cherry blossoms are more colorful, with clearer image texture. In the second row, SCALECRAFTER results show that the rabbit armor is mottled and the details of the rabbit claws are blurred, whereas DMFFT results show that the rabbit

Fig. 16. Samples generated by SCALECRAFTER with or without DMFFT.

Fig. 17. Comparison of visual effects of SD, SD + Free U, SD + DMFFT.

Fig. 18. Comparison of image quality of SCALECRAFTER, SCALECRAFTER + DMFFT.

armor is obviously repaired, the details of the claws are clearly depicted, and the generation effect achieved a 3D visual effect. It can be observed that the DMFFT approach described in this study has a wider application in the fields of visual effects enhancement and image quality improvement. The DMFFT's scaling factors are set as follows: 1. The high frequency is unchanged, the skip's low frequency amplitude scaling factor is set to 0.6. 2. The low frequency is unchanged, the backbone's high frequency amplitude scaling factor is set to 1.3.

Improve structural layout and artistic style diversity

The DMFFT method proposed in this study can also improve the diversity of generated results on the basis of text alignment and image quality improvement. Fig. 19 shows the result sets of different structures and layouts generated by DMFFT using two different texts. Each image in the first row has the same semantic content and style, but the layout, shape, and color are all different. The placement of castles, the distribution of bees, the size of sunflowers, and so on in the second row vary, but they are all aligned with the text. These rich and diversified T2I results demonstrate how DMFFT can boost structural and layout diversity.

Fig. 20 shows from the level of artistic style that DMFFT can generate rich scenes, colors and artistic expression while aligning with text. The first image in the DMFFT result set realizes the effect of Chinese style. The second

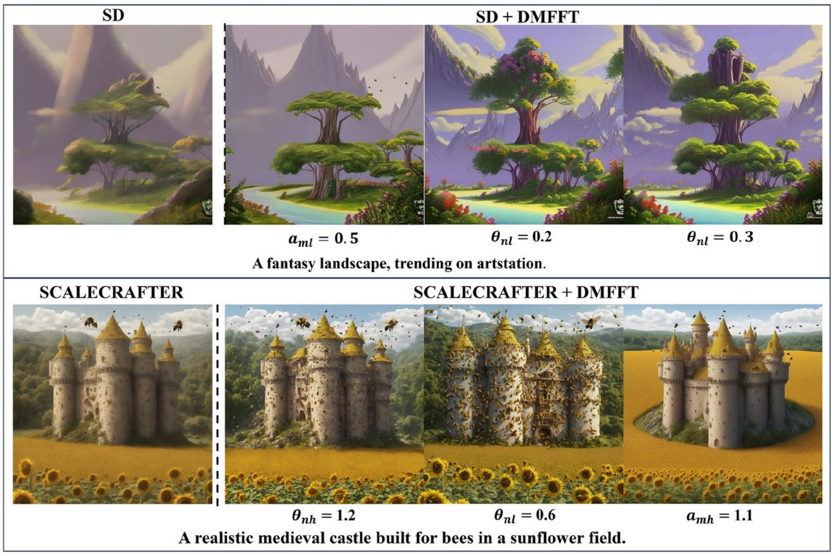


Fig. 19. DMFFT can boost the structural and layout diversity of T2I results.

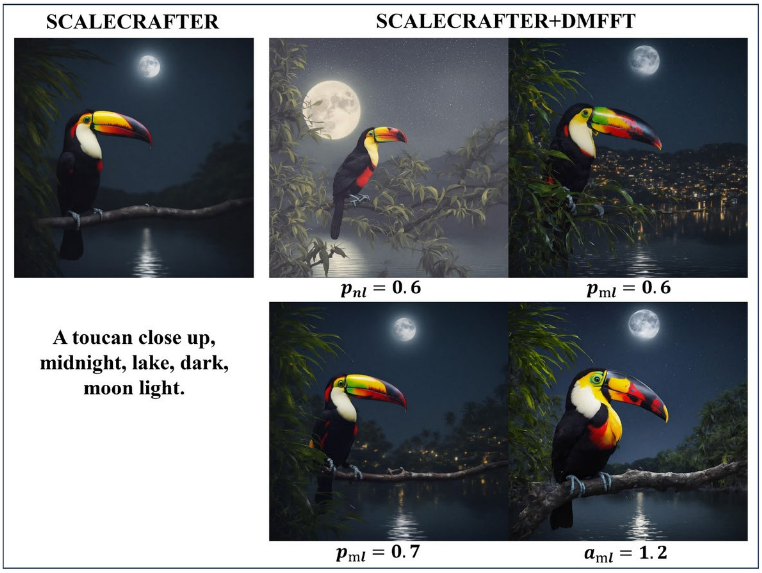


Fig. 20. DMFFT can improve the artistic style diversity of T2I results.

and third images enhance the scene generated by SD. The fourth image improves the brightness and detail of the SD generated results. To summarize, adjusting the scaling factors of high frequency, low frequency, amplitude, and phase can improve the structure, color, and artistic style, which further verifies that the proposed method combining Fourier transform and diffusion model has significant application potential in the field of T2I.

The preceding studies demonstrate DMFFT's significant enhancement over the baselines in text-to-image generation. We comprehensively evaluate the inference time of each method in many sets of improved experiments, and achieved the following comparison results: SD 1.52s, SD+FreeU 1.57s, SD+DMFFT 1.76s, SCALECRAFTER 33.35s, SCALECRAFTER+DMFFT 37.16s. It can be shown that utilizing FFT increases the inference time, but the increase time is very small, only 0.24s compared with SD, and only 3.81s compared with SCALECRAFTER. Despite the inference time is increased, DMFFT demonstrates a substantial improvement in the quality of generated images. DMFFT generates significantly higher-quality outputs, with sharper details, better text alignment, improved visual effect, and increased artistic style diversity. While the inference time has increased, the improvement in visual effect and overall quality makes the additional computational cost worthwhile.

Text-to-Video (T2V)

The proposed DMFFT method has obvious advantage in maintaining the consistency between video frames, text alignment and temporal consistency. Free-Bloom[33] and Text2Video-Zero[34] serve as baselines for further comparison and verification. The comparison results in Fig. 21 show that DMFFT can perform well in the long text guided video generation experiment, generating video results that are clearly consistent with text semantics. Such as the red text “with no visible signs of activity” and “Molten lava begins to flow down the volcano’s slopes”, the corresponding semantic expression is implemented in the first and third frames of the video generated by DMFFT, however Free-Bloom does not align the red text. In DMFFT results, the high frequency remains unchanged, while the low frequency phase of skip is scaled by 0.3.

Fig. 22 shows DMFFT’s improvement in video temporal consistency on the baseline Text2Video-Zero. The yellow box in the figure shows evident human look and body structure anomalies, whereas the red box shows obvious texture anomalies. DMFFT can improve the aforementioned abnormalities while improving the overall smoothness of the video. In the figure, the scaling factors for DMFFT generating results are set as follows: the first is 1.2 for the backbone’s high frequency, the second is high frequency unchanged, and the skip’s low frequency amplitude scaling factor is set to 0.7.

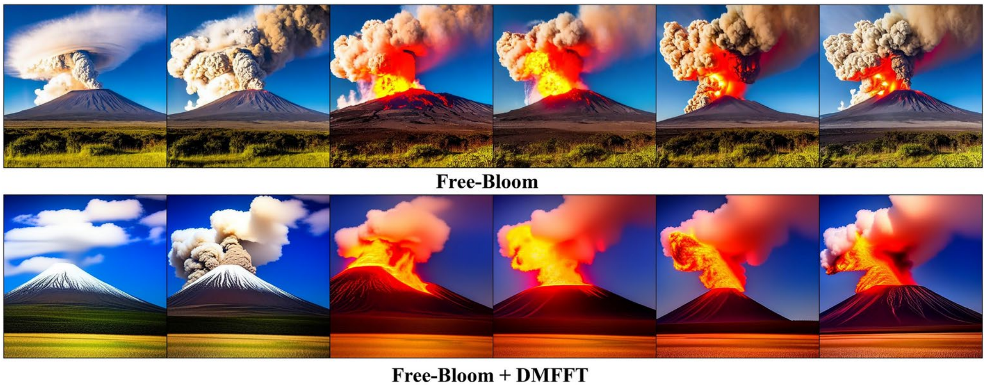
Fig. 23, based on Free-Bloom, shows the rich and diverse T2V generation results achieved by the proposed DMFFT method for the same text. The video effects presented all have high-definition visual quality, and the structural shapes of flowers and leaves vary, showing different species and growth environments. Fig. 24 shows other results of DMFFT in improving the quality of T2V generation.

In terms of text-to-video generation, the inference time of each method is as follows: Free-Bloom 10.36s, Free-Bloom+DMFFT 12.74s, Text2Video-Zero 19.94s, Text2Video-Zero+DMFFT 24.82s. The increased inference time improves the text alignment and video temporal consistency of the generated results. The higher-quality outputs make DMFFT more suitable for demanding applications, such as professional video production or high-resolution artistic content creation. In scenarios where visual quality is critical, the additional inference time is a reasonable investment for achieving superior results.

Quantitative analysis

According to the previous research[35,36], CLIP(Text) (text alignment evaluation) and CLIP(Frame) (frame consistency evaluation) are selected as the corresponding evaluation indicators. In view of the different objects in the evaluation process of T2I and T2V, CLIP (T) is refined into CLIP (Image-Text) and CLIP(Video-Text). CLIP (Image-Text) refers to the average cosine similarity between the text and the generated image. CLIP (Video-Text) refers to the average cosine similarity between all frames of the text and the output video. The two evaluation indicators refined by CLIP (T) represent the alignment degree of the generated image or video with the text guidance conditions. The larger the indicator, the better. CLIP (Frame) refers to the average cosine similarity between all consecutive frame pairs of the output video. It is used to measure the smoothness of the video generated by T2V. The larger the indicator, the better. Following the standard evaluation protocol, we measure the Frechet Inception Distance[37] (FID) between generated images and reference images to evaluate the generated image quality and diversity. To ensure fairness, we use the same randomly sampled random seed for generating images. We also measure the Frechet Video Distance[38] (FVD) between generated videos and reference videos to evaluate the generated video quality.

As shown in the quantitative analysis results in Table 2 and Table 3, the proposed method outperforms the contrasting baselines in all evaluation metrics. The CLIP(Image-Text) and CLIP(Video-Text) metrics both represent the semantic matching degree between the generated image or video and the input text, with DMFFT receiving the highest score, indicating the best consistency between the generated content and the text description. The CLIP(Frame) further reflects the temporal consistency of the generated video. After DMFFT optimization,



A towering volcano stands against a backdrop of clear blue skies, **with no visible signs of activity**. Suddenly, a plume of thick smoke and ash erupts from the volcano’ s summit, the plume billowing high into the air. **Molten lava begins to flow down the volcano's slopes**, the lava glowing brightly with intense heat and leaving a trail of destruction in its path. Explosions rock the volcano as fiery projectiles shoot into the sky, the projectiles scattering debris and ash in all directions. The eruption intensifies, with a massive column of smoke and ash ascending into the atmosphere, the column darkening the surrounding area and creating a dramatic spectacle. As the eruption reaches its peak, a pyroclastic flow cascades down the volcano's sides, the flow engulfing everything in its path with a deadly combination of hot gases, ash, and volcanic material.

Fig. 21. Comparison of video text alignment of Free-Bloom and Free-Bloom + DMFFT.

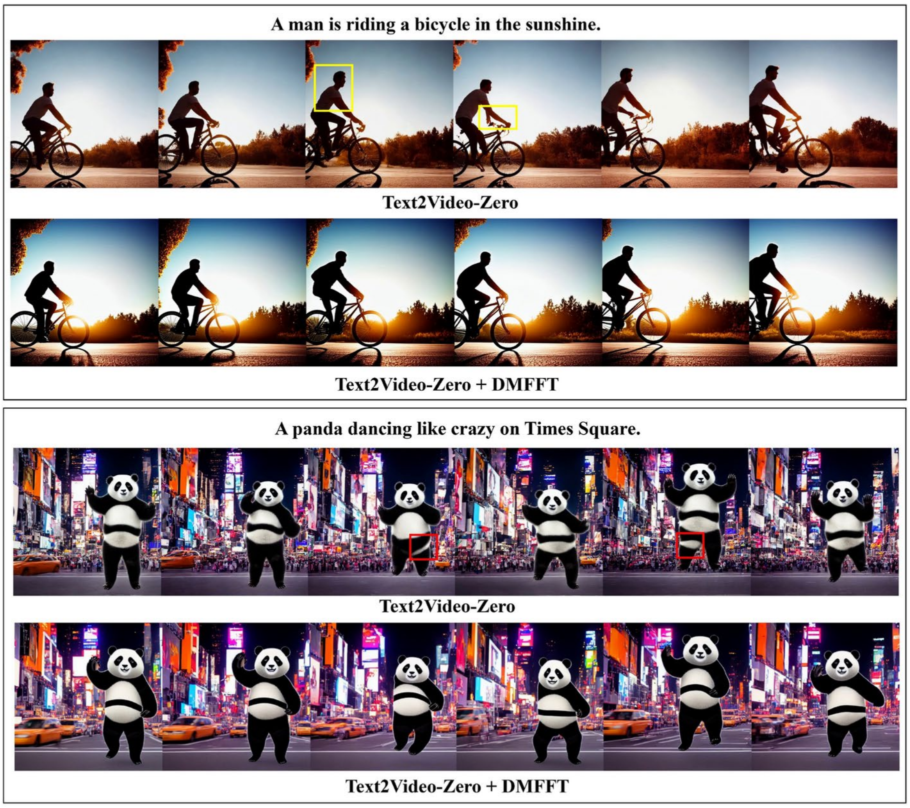


Fig. 22. Comparison of video temporal consistency of Text2Video-Zero and Text2Video-Zero + DMFFT.

Fig. 23. Multiple results of Free-Bloom + DMFFT based on the same text.

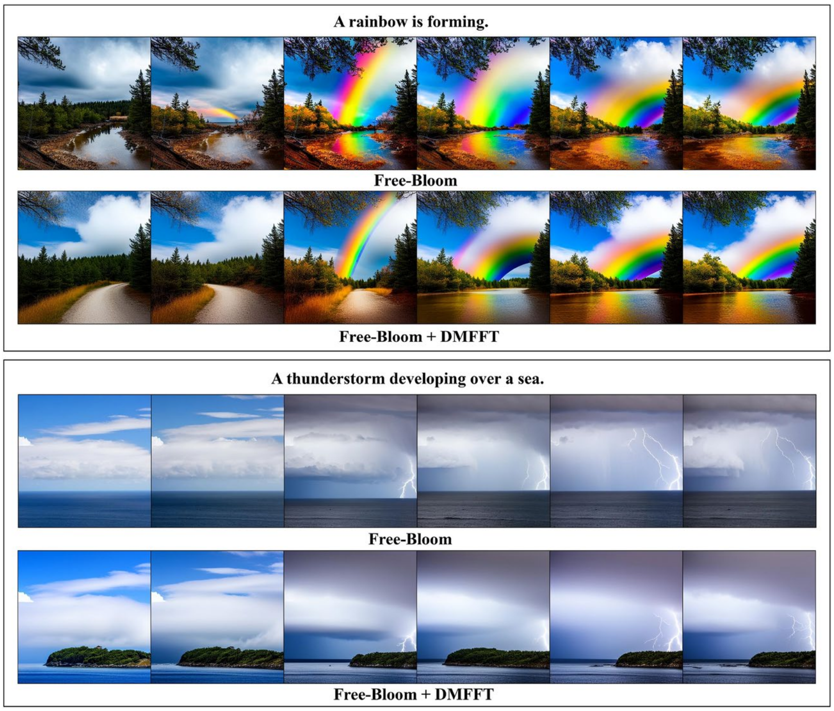


Fig. 24. Other T2V quality improvement results of DMFFT.

Method	CLIP (Image-Text) ↑	FID ↓
SD[5]	0.278	64.80
SD + FreeU[27]	0.277	65.94
SD + DMFFT	**0.291**	**38.50**
SCALECRAFTER[32]	0.309	69.50
SCALECRAFTER + DMFFT	**0.310**	**32.67**

Table 2. Quantitative comparison of T2I generation quality. Significant values are in bold.

Method	CLIP (Video-Text) ↑	CLIP (Frame) ↑	FVD ↓
Text2Video-Zero[34]	0.285	0.970	1355.29
Text2Video-Zero + DMFFT	**0.289**	**0.972**	**674.14**
Free-Bloom[33]	0.251	0.944	494.08
Free-Bloom + DMFFT	**0.257**	**0.950**	**139.00**

Table 3. Quantitative comparison of T2V generation quality. Significant values are in bold.

T2V video results not only achieve text alignment, but also further improve video temporal consistency and smooth performance. When compared to the CLIP series evaluation metrics, the proposed method significantly outperforms other comparative baselines in the FID and FVD metrics. We conclude that this is mostly owing to the fact that FID and FVD metrics are more sensitive to the content diversity and quality of generated images or videos, as well as being better at capturing the spatiotemporal consistency of videos.

Ablation study

To validate the effectiveness of each frequency domain component, we conduct ablation studies in qualitative way. As shown in FIG. 25, significant visual effects can be obtained by simply applying a scaling factor. Through comparison, it is discovered that some results generated by high frequency components (skip HF amplitude, backbone HF amplitude, skip HF, etc.) have bright colors. Most of the frequency domain components in the skip branch (skip HF, skip LF amplitude, skip LF phase, etc.) make the generated results have clearer textures and richer image semantics. Most of the high-frequency components of the backbone branch make the visual background appear more dreamlike. The high frequency or low frequency components and the corresponding

Fig. 25. Only one scaling factor is applied. HF represents high frequency, and LF represents low frequency. According to the previous analysis, the high frequency phase of the backbone has little influence on the generated results, so the original image is placed in the corresponding position, marked as unchanged.

high frequency amplitude or low frequency amplitude of each branch generate image content with similar semantic style (such as skip HF amplitude and skip HF, skip LF amplitude and skip LF, etc.). Individual phase components can generate results that differ significantly from the others (e.g., skip LF phase).

We further verify the influence of superposing the scaling factors of two frequency domain components on the generated results, as shown in FIG. 26. The first group contains the results obtained by applying the scaling factors of skip high frequency amplitude and skip high frequency phase simultaneously. The second group applied both skip high frequency and skip low frequency, and the third group applied both backbone low-frequency amplitude and skip low frequency amplitude. Three groups of superposition experiments are representative. The generation effect after superposition reveals that the results obtained after superimposing two scaling factors are inconsistent with the image before superposition, and the semantic style, image quality, and image style of the generated results have changed, making them uncertain. Therefore, it is suggested that users should mainly apply a single scaling factor in the use process, and some individual scaling factors can be superimposed appropriately.

Discussion

The proposed method improves the generation quality of diffusion models and generates visually significant results. However, by improving the visual effect, the fidelity of some experimental results is compromised. For example, in Fig. 17 and Fig. 18, the DMFFT-generated results are relatively high resolution and have excellent text alignment, but they exhibit the movie and television special effects or modeling effects. Furthermore, in Fig. 21, DMFFT achieves temporal consistency and text alignment, but the generated video lacks the variation of the real scene in which the grass scorches, as shown in Free-Bloom. Through analysis, it is found that the above examples respectively adjust the scaling factor of the high-frequency component of the backbone branch, the high-frequency amplitude component of the backbone branch, and the low-frequency phase component of the skip branch. In the method section, Fig. 8, Fig. 11, and Fig. 14 demonstrate that adjusting these components can cause the generated results to exhibit the movie and television special effects or modeling effects. In addition, sampling different random seeds will have a certain impact on the generation effect. To further improve the realism of the generated effect, we can adjust the scaling factor of other frequency domain components, or the random seeds can be resampled, so as to achieve both realistic and significant visual effects.

The advancement of science and technology has promoted the development of various generative models. Text-to-image and text-to-video are representative applications of generative artificial intelligence, which have received extensive attention. As artificial intelligence researchers, we must consider the appropriate application of this technology when learning and using it. Avoid using the model to generate data that violate social standards and refrain from distributing negative social values. Enhancing the ethical implications of T2I and T2V generation techniques requires a collaborative effort. Through technological improvement, legal regulation, and public education, it is possible to enjoy the technological dividend while minimizing its potential negative impact.

Conclusions

In this study, we propose a novel optimization method to improve the generation quality of Diffusion Models with Fourier transform, called DMFFT, which is completely based on the baselines without additional computational costs. DMFFT can be easily integrated into existing diffusion models and can effectively improve the textual

Fig. 26. Two scaling factors are applied.

semantics, structure, color, variety, and artistry of generated results by only selectively adjusting frequency or amplitude or phase scaling factors during inference. The proposed DMFFT can be directly integrated into diffusion model-based T2I or T2V studies without training, and can meet most quality improvement needs, which is a general-purpose method that can be widely applied. The significant performance of DMFFT on image and video generation tasks show that our method has significant applications in art, film and television production, 3D modeling, and other fields.

Data availibility

The datasets used and analysed during the current study available from the corresponding author on reasonable request.

Code availability

The readers can access the code for this study through the following repository link. There are no corresponding access restrictions. We will maintain the code in the repository. DOI: https://doi.org/10.5281/zenodo.14838639.

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References

1. Ho, J., Jain, A. & Abbeel, P. Denoising diffusion probabilistic models. In *Proceedings of the 34th International Conference on Neural Information Processing Systems*, NIPS '20. (Curran Associates Inc., Red Hook, NY, USA, 2020).
2. Sohl-Dickstein, J., Weiss, E. A., Maheswaranathan, N. & Ganguli, S. Deep unsupervised learning using nonequilibrium thermodynamics. In *Proceedings of the 32nd International Conference on International Conference on Machine Learning - Volume 37*, ICML'15, 2256–2265 (JMLR.org, 2015).

3. Song, Y. & Ermon, S. Generative modeling by estimating gradients of the data distribution. In Wallach, H. M. *et al.* (eds.) *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8-14, 2019, Vancouver, BC, Canada*, 11895–11907 (2019).
4. Song, Y. *et al.* Score-based generative modeling through stochastic differential equations. In *9th International Conference on Learning Representations, ICLR 2021, Virtual Event, Austria, May 3-7, 2021* (OpenReview.net, 2021).
5. Rombach, R., Blattmann, A., Lorenz, D., Esser, P. & Ommer, B. High-resolution image synthesis with latent diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2022, New Orleans, LA, USA, June 18-24, 2022*, 10674–10685, https://doi.org/10.1109/CVPR52688.2022.01042 (IEEE, 2022).
6. Yang, L. et al. Diffusion models: A comprehensive survey of methods and applications. *ACM Comput. Surv.* **56**, 1–39 (2022).
7. Saharia, C. *et al.* Photorealistic text-to-image diffusion models with deep language understanding. In Koyejo, S. *et al.* (eds.) *Advances in Neural Information Processing Systems 35: Annual Conference on Neural Information Processing Systems 2022, NeurIPS 2022, New Orleans, LA, USA, November 28 - December 9, 2022* (2022).
8. Karras, T., Aittala, M., Aila, T. & Laine, S. Elucidating the design space of diffusion-based generative models. In Koyejo, S. *et al.* (eds.) *Advances in Neural Information Processing Systems 35: Annual Conference on Neural Information Processing Systems 2022, NeurIPS 2022, New Orleans, LA, USA, November 28 - December 9, 2022* (2022).
9. Schuhmann, C. *et al.* Laion-5b: An open large-scale dataset for training next generation image-text models. In Koyejo, S. *et al.* (eds.) *Advances in Neural Information Processing Systems*, vol. 35, 25278–25294 (Curran Associates, Inc., 2022).
10. Zhang, L., Rao, A. & Agrawala, M. Adding conditional control to text-to-image diffusion models. In *IEEE/CVF International Conference on Computer Vision, ICCV 2023, Paris, France, October 1-6, 2023*, 3813–3824, https://doi.org/10.1109/ICCV51070.2023.00355 (IEEE, 2023).
11. Mou, C. *et al.* T2i-adapter: Learning adapters to dig out more controllable ability for text-to-image diffusion models. In Wooldridge, M. J., Dy, J. G. & Natarajan, S. (eds.) *Thirty-Eighth AAAI Conference on Artificial Intelligence, AAAI 2024, Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence, IAAI 2024, Fourteenth Symposium on Educational Advances in Artificial Intelligence, EAAI 2014, February 20-27, 2024, Vancouver, Canada*, 4296–4304, https://doi.org/10.1609/AAAI.V38I5.28226 (AAAI Press, 2024).
12. Ruiz, N. *et al.* Dreambooth: Fine tuning text-to-image diffusion models for subject-driven generation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2023, Vancouver, BC, Canada, June 17-24, 2023*, 22500–22510, https://doi.org/10.1109/CVPR52729.2023.02155 (IEEE, 2023).
13. Brooks, T., Holynski, A. & Efros, A. A. Instructpix2pix: Learning to follow image editing instructions. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2023, Vancouver, BC, Canada, June 17-24, 2023*, 18392–18402, https://doi.org/10.1109/CVPR52729.2023.01764 (IEEE, 2023).
14. Sohn, K. *et al.* Styledrop: Text-to-image generation in any style. Preprint at arXiv:2306.00983 (2023).
15. Xie, S., Zhang, Z., Lin, Z., Hinz, T. & Zhang, K. Smartbrush: Text and shape guided object inpainting with diffusion model. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2023, Vancouver, BC, Canada, June 17-24, 2023*, 22428–22437, https://doi.org/10.1109/CVPR52729.2023.02148 (IEEE, 2023).
16. Pan, X. *et al.* Drag your GAN: interactive point-based manipulation on the generative image manifold. In Brunvand, E., Sheffer, A. & Wimmer, M. (eds.) *ACM SIGGRAPH 2023 Conference Proceedings, SIGGRAPH 2023, Los Angeles, CA, USA, August 6-10, 2023*, 78:1–78:11, https://doi.org/10.1145/3588432.3591500 (ACM, 2023).
17. Hong, S., Lee, G., Jang, W. & Kim, S. Improving sample quality of diffusion models using self-attention guidance. In *IEEE/CVF International Conference on Computer Vision, ICCV 2023, Paris, France, October 1-6, 2023*, 7428–7437, https://doi.org/10.1109/ICCV51070.2023.00686 (IEEE, 2023).
18. Feng, Z. *et al.* Ernie-vilg 2.0: Improving text-to-image diffusion model with knowledge-enhanced mixture-of-denoising-experts. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2023, Vancouver, BC, Canada, June 17-24, 2023*, 10135–10145, https://doi.org/10.1109/CVPR52729.2023.00977 (IEEE, 2023).
19. Podell, D. *et al.* Sdxl: Improving latent diffusion models for high-resolution image synthesis. Preprint at arXiv:abs/2307.01952 (2023).
20. Lee, S. *et al.* Priorgrad: Improving conditional denoising diffusion models with data-dependent adaptive prior. In *The Tenth International Conference on Learning Representations, ICLR 2022, Virtual Event, April 25-29, 2022* (OpenReview.net, 2022).
21. Yang, Y. & Soatto, S. FDA: fourier domain adaptation for semantic segmentation. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2020, Seattle, WA, USA, June 13-19, 2020*, 4084–4094, https://doi.org/10.1109/CVPR42600.2020.00414 (Computer Vision Foundation / IEEE, 2020).
22. Suvorov, R. *et al.* Resolution-robust large mask inpainting with fourier convolutions. In *IEEE/CVF Winter Conference on Applications of Computer Vision, WACV 2022, Waikoloa, HI, USA, January 3-8, 2022*, 3172–3182, https://doi.org/10.1109/WACV51458.2022.00323 (IEEE, 2022).
23. Zhu, Y. *et al.* Fourier contour embedding for arbitrary-shaped text detection. In *IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2021, virtual, June 19-25, 2021*, 3123–3131, https://doi.org/10.1109/CVPR46437.2021.00314 (Computer Vision Foundation / IEEE, 2021).
24. Lee-Thorp, J., Ainslie, J., Eckstein, I. & Ontañón, S. Fnet: Mixing tokens with fourier transforms. In Carpuat, M., de Marneffe, M. & Ruíz, I. V. M. (eds.) *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL 2022, Seattle, WA, United States, July 10-15, 2022*, 4296–4313, https://doi.org/10.18653/V1/2022.NAACL-MAIN.319 (Association for Computational Linguistics, 2022).
25. Buchholz, T. & Jug, F. Fourier image transformer. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, CVPR Workshops 2022, New Orleans, LA, USA, June 19-20, 2022*, 1845–1853, https://doi.org/10.1109/CVPRW56347.2022.00201 (IEEE, 2022).
26. Mao, X. *et al.* Intriguing findings of frequency selection for image deblurring. In Williams, B., Chen, Y. & Neville, J. (eds.) *Thirty-Seventh AAAI Conference on Artificial Intelligence, AAAI 2023, Thirty-Fifth Conference on Innovative Applications of Artificial Intelligence, IAAI 2023, Thirteenth Symposium on Educational Advances in Artificial Intelligence, EAAI 2023, Washington, DC, USA, February 7-14, 2023*, 1905–1913, https://doi.org/10.1609/AAAI.V37I2.25281 (AAAI Press, 2023).
27. Si, C., Huang, Z., Jiang, Y. & Liu, Z. Freeu: Free lunch in diffusion u-net. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2024, Seattle, WA, USA, June 16-22, 2024*, 4733–4743, https://doi.org/10.1109/CVPR52733.2024.00453 (IEEE, 2024).
28. Ronneberger, O., Fischer, P. & Brox, T. U-net: Convolutional networks for biomedical image segmentation. In Navab, N., Hornegger, J., III, W. M. W. & Frangi, A. F. (eds.) *Medical Image Computing and Computer-Assisted Intervention - MICCAI 2015 - 18th International Conference Munich, Germany, October 5 - 9, 2015, Proceedings, Part III*, vol. 9351 of *Lecture Notes in Computer Science*, 234–241, https://doi.org/10.1007/978-3-319-24574-4_28 (Springer, 2015).
29. Zeiler, M. D. & Fergus, R. Visualizing and understanding convolutional networks. In Fleet, D. J., Pajdla, T., Schiele, B. & Tuytelaars, T. (eds.) *Computer Vision - ECCV 2014 - 13th European Conference, Zurich, Switzerland, September 6-12, 2014, Proceedings, Part I*, vol. 8689 of *Lecture Notes in Computer Science*, 818–833, https://doi.org/10.1007/978-3-319-10590-1_53 (Springer, 2014).
30. Nair, V., Chatterjee, M., Tavakoli, N., Namin, A. S. & Snoeyink, C. Fast fourier transformation for optimizing convolutional neural networks in object recognition. Preprint at arXiv:2010.04257 (2020).
31. Han, Y. & Hong, B.-W. Deep learning based on fourier convolutional neural network incorporating random kernels. *Electronics* (2021).

32. He, Y. *et al.* Scalecrafter: Tuning-free higher-resolution visual generation with diffusion models. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024* (OpenReview.net, 2024).
33. Huang, H. *et al.* Free-bloom: Zero-shot text-to-video generator with LLM director and LDM animator. In Oh, A. *et al.* (eds.) *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023* (2023).
34. Khachatryan, L. *et al.* Text2video-zero: Text-to-image diffusion models are zero-shot video generators. In *IEEE/CVF International Conference on Computer Vision, ICCV 2023, Paris, France, October 1-6, 2023*, 15908–15918, https://doi.org/10.1109/ICCV51070.2023.01462 (IEEE, 2023).
35. Wu, J. Z. *et al.* Tune-a-video: One-shot tuning of image diffusion models for text-to-video generation. In *IEEE/CVF International Conference on Computer Vision, ICCV 2023, Paris, France, October 1-6, 2023*, 7589–7599, https://doi.org/10.1109/ICCV51070.2023.00701 (IEEE, 2023).
36. Zhao, M., Wang, R., Bao, F., Li, C. & Zhu, J. Controlvideo: Adding conditional control for one shot text-to-video editing. Preprint at arXiv:2305.17098 (2023).
37. Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B. & Hochreiter, S. Gans trained by a two time-scale update rule converge to a local nash equilibrium. In: Guyon, I. *et al.* (eds.) *Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems 2017, December 4-9, 2017, Long Beach, CA, USA*, 6626–6637 (2017).
38. Unterthiner, T. *et al.* Towards accurate generative models of video: A new metric & challenges. *CoRR***abs/1812.01717**. arXiv:1812.01717. (2018).

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Declarations

Competing interests
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Experimental results statement
All experimental results are guided by text, so all the results are virtual, including the people in the results are also virtual character, not real.

Additional information
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